

1. The $ETS(AAdN)$ model is given by the system

$$\begin{cases} u_t \sim \mathcal{N}(0; 16) \\ b_t = 0.5b_{t-1} + 0.8u_t \\ \ell_t = \ell_{t-1} + 0.5b_{t-1} + 0.8u_t \\ y_t = \ell_{t-1} + 0.5b_{t-1} + u_t \end{cases}$$

with $\ell_{100} = 30$ and $b_{100} = 2$.

- (a) [5] Find the 95% predictive interval for y_{102} .
 - (b) [5] Find conditional probability $\mathbb{P}(y_{102} < 40 \mid \ell_{100}, b_{100})$ in terms of the standard normal distribution function.
2. The semi-annual y_t is modelled by $ETS(AAA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 16) \\ b_t = b_{t-1} + 0.8u_t \\ s_t = s_{t-2} + 0.1u_t \\ \ell_t = \ell_{t-1} + b_{t-1} + 0.5u_t \\ y_t = \ell_{t-1} + b_{t-1} + s_{t-2} + u_t \end{cases}$$

with $\ell_0 = 20$, $b_0 = 1$, $s_0 = 5$ and $s_{-1} = -5$.

Decompose first two observations $y_1 = 25$ and $y_2 = 20$ into trend, seasonal and error components.

3. Consider $MA(1)$ process $y_t = 3 + u_t + 4u_{t-1}$, where u_t is a white noise process with variance 10.
- (a) [5] Find $\mathbb{E}(y_t)$, $\text{Var}(y_t)$, $\text{Cov}(y_t, y_s)$.
 - (b) [5] Consider an independent copy (y'_t) of (y_t) (all expected value, variance and autocovariances are the same). If possible classify $s_t = y'_t + y_{t-1}$ as $MA(q)$ process.
4. Consider the equation $y_t = 0.5y_{t-1} + u_t + 0.4u_{t-1}$, where u_t is a white noise process with variance σ^2 .
- (a) [1] How many non-stationary solutions does this equation have?
 - (b) [3] How many stationary solutions of $MA(\infty)$ form with respect to (u_t) does this equation have? Find all solutions, if possible.
 - (c) [2] Rewrite all stationary solutions in $AR(\infty)$ form with respect to (u_t) , if possible.
 - (d) [4] Find $\text{Var}(y_t)$ for all stationary solutions.
5. Consider the stationary solution of the equation

$$y_t = 0.8y_{t-1} - 0.5y_{t-2} + u_t,$$

where (u_t) is a white noise.

- (a) [3] Check the existence of stationary solution.
 - (b) [4] Find the values of ACF ρ_1, ρ_2, ρ_3 .
 - (c) [3] Find the values of PACF $\phi_{11}, \phi_{22}, \phi_{33}$.
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6. You know that y_t is a stationary $ARMA(p, q)$ process.
- (a) Is it possible that autocorrelation $\rho_{10} = 0.5$ but all other autocorrelations (ρ_k for $k \geq 1$ and $k \neq 10$) are zero? If possible provide an equation of such a process.
 - (b) Is it possible that partial autocorrelation $\phi_{10,10} = 0.5$ but all other partial autocorrelations ($\phi_{k,k}$ for $k \geq 1$ and $k \neq 10$) are zero? If possible provide an equation of such a process.
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